CSMI17 – Artificial Intelligence

Assignment

1. **Robot Path-finding Problem**: In a grid, a robot is initially located in a cell and must find its way to the goal located in another cell. There may be obstacles in the grid which the robot has to avoid.
   Use A* search to solve the robot path-finding problem. Find at least 3 heuristics for A* search and compare the performance of the heuristics.
   (*You need to first develop the code for creating a grid with random obstacles, random starting cell and random goal cell. Then, develop the code for each heuristic in A* search. For each heuristic, run the program multiple times with randomly generated grids to get the performance of each heuristic. Then create a graph to compare the heuristics. You must decide the metrics with which the heuristics can be compared.*)

2. **Multi-Robot Path Finding Problem**: Now consider the same problem as above but with multiple robots. Each robot has its own starting point and a goal to reach. You can try the A* algorithm for this setting.
   What are the additional challenges encountered now? Propose a new method using which some of these additional challenges can be solved. Experiment with multiple configurations of obstacles, multiple grid sizes and different number of agents. Show how your method works better than plain A* using analytics similar to the previous question.

For both of the above problems, you must finish the coding in any language of your choice, upload in GitHub and submit a report containing:

1. Problem Definition
2. Assumptions/customizations
3. Description of the algorithms (or heuristics)
4. Experimental setup – Description and screenshots
5. Performance comparison
6. GitHub link

The project can be done in teams of 2 or 3 students each. You have to present your work in class.

Weightage of marks:

Q1 – 5

Q2 – 20 (Identification of challenges – 5; Novelty of the proposed solution – 15)

Implementation – 10

Report – 5

Presentation – 10